

Termwise equivalence of CWY2 and wY: paper proof

DEFINITIONS & PROOFS / ENGLISH EDITION

Written 2026-09-16; repository edition 2026-09-17. The systems are **current CWY2 without special columns** and **omega-Y weak magma**. This is not CWY2 legacy, weak-omega-Y or the class of arbitrary free ancestor-word tables.

0. Statement, domain and dependencies

Let F_n be default wY expansion and f_n the mathematical algorithm of `cwy_direct.mjs`, disregarding browser time, output, machine-integer and input guards. The external tops use $(1, n+1)$ and E_{n+1} as their n th terms.

For the empty string and all finite positive-integer strings s starting with 1, the highest-profile encoding satisfies

$$f_n(\Phi(s)) = \Phi(F_n(s)), \quad s <_{\text{lex}} t \iff \Phi(s) <_{\text{collex}} \Phi(t). \quad (0.1)$$

It also preserves width, prefixes, zero and successors. On standard domains generated by corresponding seeds and operations it is an onto order isomorphism, not just a one-way embedding. Their order types agree, also after adjoining their respective external tops.

Dependency boundary. We use the following finite combinatorial results of the supplied manuscript [W]: Definitions 1.2-1.3, Lemmas 2.1, 2.3 and 2.4, Proposition 3.6, Reconstruction Theorem 4.8, Definition 5.4, Theorem 5.5 and Anchor Lemma 6.1. Sections 2-4 supply the additional slice-lifting, intermediate-level and origin-query arguments. The global wY well-ordering theorem is not a premise of the local identities.

This is an existing **paper proof relative to those lemmas**, not a Lean certificate, a complete re-formalization of [W], or a new independent audit of all its results. Finite testing is not a logical premise. Neither the definition nor the algorithm changed for this proof.

1. Encoding and notation

Draw the canonical mountain $M(s)$. Following [W], let $x(e)$, $h(e)$ be a node's column and height, $\ell(e)$, $r(e)$ its left and right feet, and $c(e)$ its immediate cap. For a non-bottom node let $d(e)$ be its least jump dimension and $P(e) = c(\ell(e))$ its parent cap. Put $H(v) = \ell(c(v))$ when the cap exists. Let $I(e)$ be the normalized profile word of Definition 5.4, ordered by $<_*$.

Bottom nodes have no word: assign them the auxiliary symbol $\perp$, below all actual words. $L_{a \rightarrow b}$ replaces address a by b in a word. For height inequalities only, regard a missing cap as $+\infty$; it is not an actual node or a word-bearing object.

Column C_j of $\Phi(s)$ retains the greatest word for each left-foot column $p(e) = x(\ell(e))$, sorted by decreasing p . This is a **column address**, not the row address of a parent cap.

Moving upward in a column, parent addresses do not increase: the next search starts in the current left-foot column and can only move left. Together with Theorem 5.5's strictly increasing words, each parent occupies a consecutive group. The saved records are exactly the group tops, with words increasing in storage order. They satisfy CWY2's skyline form.

Sky means: take the greatest word for each parent, then discard rows dominated by an at-least-as-large word at a larger parent. It is the finite Pareto frontier of the two coordinates (parent, word).

2. Slice-ancestor lifting

Fix a height $u \geq 1$ and let $q_i(u)$ be the highest node of column i at or below u . The slice parent edge is $i \rightarrow x(H(q_i(u)))$ when defined. Lemma 2.1 of [W] says this points to the parent column's snapshot at the same slice and does not increase height. Lemma 2.4 says ancestry at a higher slice persists at a lower slice.

Lemma 2.1 (exact lifting of low nodes). Suppose p is a strict ancestor of j at slice u . For every non-bottom node a in column p of height $\mu \leq u$, there is a unique node a' in column j such that

$$h(a') = \mu, \quad d(a') = d(a), \quad I(a') = L_{p \rightarrow j}(I(a)), \quad p(a') \geq p. \quad (2.1)$$

The same-height statement also holds for bottom nodes, without assigning them words.

Proof. The mountain is finite. Let v be its greatest actual height strictly below μ . Since a is not bottom, $1 \leq v < \mu \leq u$. Inherited ancestry gives a path from j to p at slice v . The terminal snapshot's cap is exactly a .

The snapshot caps along that path form a P-chain. By the parent-cap bound their heights do not decrease toward p ; every one is above v and at most the terminal height μ . There is no actual height strictly between v and μ , so all these caps have height μ . In particular column j has such a first cap a' .

Equation (2.2) of [W] makes adjacent same-height parent caps have equal dimension, hence the whole chain has dimension $d(a)$. The word recurrence repeatedly replaces the same number of trailing letters by the intermediate column address, finally giving (2.1). The first parent along the path is at least p . Uniqueness follows from strictly increasing heights in each column. Bottom existence at height 1 comes from the common bottom scaffold. This argument stays inside the original mountain and assumes no correctness of CWY2 decrement. QED.

3. Lowering a positive-dimensional controlling word

Let e be a non-bottom node in column j , $f = r(e)$, $L = \ell(e)$, $p = x(L)$, $\lambda = h(L)$, $B = h(f)$ and $d = d(e) > 0$. Put $U = I(P(e))$, or $U = (p)$ if the parent cap is absent.

The parent-cap bound and word recurrence say that the last d padded coordinates of U are all p and

$$I(e) = T_{d,j}^*(U).$$

Let θ be the normalization of $I(e)$ after its first j is replaced by p . Then

$$\theta = T_{d-1,j}^*(U). \quad (3.1)$$

Lemma 3.1. The node f is non-bottom and

$$I(f) \geq_* \theta. \quad (3.2)$$

If f has the same parent column p , equality holds.

3.1 The mandatory intermediate level

Put $k = d - 1$. Least-jump minimality gives

$$\lambda < \mu := \lambda + \omega^k \leq B < \lambda + \omega^{k+1} = h(e). \quad (3.3)$$

Thus $B > 1$ and f is non-bottom. At slice B , p is j 's direct parent. Lemma 2.1 gives a node of height λ in column j . Let v be the first node there with height at least μ ; it exists since f is at B . Write $\beta = h(r(v))$. Then $\lambda \leq \beta < \mu \leq h(v) \leq B$.

Ancestry inheritance keeps p an ancestor of j at slice β . Its snapshot is still L , whose cap is above B . Heights do not increase along the slice path, so v 's left-foot height lies in $[\lambda, \beta]$.

The jump dimension of v cannot be below k : a lower-power jump from below $\lambda + \omega^k$ cannot reach that height. It cannot exceed k either, since its height would then be at least $\lambda + \omega^{k+1} > B$. Therefore

$$d(v) = k, \quad h(v) = \lambda + \omega^k = \mu. \quad (3.4)$$

For $k = 0$, the interval $[\lambda, \lambda + 1)$ contains only λ , so the argument still applies.

Along the slice- β path from j to p , the snapshot caps form a P-chain, starting at v and ending at $P(e)$, or at capless L . At a common padded length, each intermediate column only replaces some final coordinates by larger column addresses. These are coordinatewise nondecreasing changes; finally k coordinates are replaced in column j . Consequently

$$I(v) \geq_* T_{k,j}^*(U) = \theta.$$

Since v is no higher than f , columnwise strict increase gives $I(f) \geq_* I(v)$, proving (3.2). The witness v is essential: domination is not inferred merely from an expectation that a virtual row should disappear.

3.2 Equality for the same parent

Suppose f 's left foot Q is also in column p . L is the highest point there at or below B , so $h(Q) \leq \lambda$. If Q were strictly below L , the parent-cap bound would yield

$$B = h(f) \leq h(c(Q)) \leq \lambda \leq B,$$

contradicting $d > 0$. Thus $Q=L$. If $d(f) = a$, then $B = \lambda + \omega^a$ and the next least jump has $d = a + 1$. Both f and e use the same parent-cap word U , giving $I(f) = T_{a-1,j}^*(U) = \theta$. QED.

4. The origin query returns the required lower word

First let $W = I(e)$ have last letter equal to its own column q . Then $d(e) > 0$. In C_q , the first group-maximum word at least W has exactly e 's parent address r : earlier groups lie below e , and later groups above it.

Let U_- be the previous group's top word, or $\perp$. Replace the first q in W by r and normalize to obtain θ . Lemma 3.1 gives two cases:

- If e and its right foot share a parent, $I(r(e)) = \theta$ and $U_- < \theta$.
- If they change parent, $r(e)$ is the previous group's top, so $I(r(e)) = U_- \geq \theta$.

Hence

$$I(r(e)) = \max_*(U_-, \theta). \quad (4.1)$$

This is precisely the single-scan belowOrigin formula. It still works if e is not a group top: the stored maximum may exceed W , but identifies the same group.

4.1 Collapsing zero-dimensional inheritance

Now take any non-bottom node e in column j . Set $W = I(e)$, q its last letter, $f = r(e)$ and $B = h(f)$. If $q = j$, the preceding argument applies. If $q < j$, then $d(e) = 0$, and e 's left foot also has height B .

Follow P while parent caps exist and the dimension is zero. The word stays W, the right-foot height stays B, and the column decreases strictly. The process terminates finitely:

- If it reaches a positive-dimensional node a, that node's column is q, its word is W and $h(r(a)) = B$. At slice B, q is an ancestor of j. Lemma 2.1 gives $I(f) = L_{q \rightarrow j}(I(r(a)))$, and (4.1) is exactly the query for $I(r(a))$.
- If it ends at a capless zero-dimensional node, $W = (q)$ and the top of column q has height B. Every word in C_q is below the singleton (q), so the scan returns that column's greatest word, or $\perp$ if there is no non-bottom node. Lifting by Lemma 2.1 gives precisely I(f), including the bottom case.

Every zero-dimensional P step is a slice parent edge between equal-height snapshots; the ancestry used here is not an extra hypothesis. In all cases,

$$L_{q \rightarrow j}(\text{belowOrigin}(\Phi(s), I(e))) = I(r(e)), \quad (4.2)$$

with $L(\perp) = \perp$. The implementation does not traverse this P-chain: the last letter q directly identifies its endpoint column.

5. Direct decrement identity

5.1 Removing the original top

Let e be the last column's top, with controlling row (p, w), and let $f = r(e)$. After deleting e:

- If f is still in the p-group, its new greatest word is I(f).
- If the parent changes, the p-group disappears and the preceding group's saved word already equals I(f).
- If f is bottom, no non-bottom node remains.

Equation (4.2) computes exactly I(f). In the changing-parent case the temporarily inserted (p, I(f)) is dominated by an existing (r, I(f)) with $r > p$, and Sky deletes it. Other groups remain unchanged. Thus `removeOriginalTop` exactly represents the original lower part.

5.2 Why the entire parent table can be merged

Lemma 2.3 of [W] preserves the lower part and grafts L's complete upper tail after numeric decrement. The implementation merges all of C_p , so one must exclude extra constraints.

At slice $B = h(f)$, p is j's parent and c(L) lies above B. Take any non-bottom node a at or below L in column p, with parent $r < p$. Lemma 2.1 gives a same-height node a' in the retained lower part of column j, with

$$I(a') = L_{p \rightarrow j}(I(a)), \quad p(a') \geq p > r. \quad (5.1)$$

The extra low row (r, L(I(a))) is therefore dominated by the actual lower part. Higher parent-column nodes are exactly the real grafted tail: their heights, left feet and dimensions are unchanged, and their words merely replace the local address p by j.

Any parent group meeting the upper tail has its maximum in that tail, so keeping only group maxima loses no new highest word. Taking same-parent maxima and then using (5.1), Sky gives exactly the highest-profile table of the true decremented mountain.

Direct decrement theorem. For each raw finite source string s whose last value exceeds 1,

$$\text{minusColumn}(\Phi(s)) = \Phi(s^-)_{\text{last}}, \quad (5.2)$$

where s^- decreases only its last value by one. This supplies the four formerly missing local obligations: same-parent lowering, changed-parent domination, zero-dimensional origin inheritance and clipping the irrelevant low parent part. No mountain reconstruction is used to define or execute the CWY2 rule.

6. Copying is exact relocation, not just an upper bound

We spell out the earlier exact-copy argument rather than replacing the inequality of [W, Lemma 6.2] by equality. The source is the repaired mountain $T = M(s^-)$. Fix root column c , last column j and width $w = j - c$. In block b let $\tau(i) = i$ for $i < c$ and $\tau(i) = i + bw$ otherwise.

Proposition 3.6(ii) of [W] identifies both feet of every ordinary copied node e with the effective source-foot images ψ .

Non-root bad parent. If $P(e)$ exists, apply that proposition to it as well: its ordinary copy has right foot $\psi(\ell(e))$. Thus the target left foot's immediate cap is exactly the ordinary parent-cap copy, not just a height upper bound. If $P(e)$ is absent, the source left foot is a column top, so cannot be a marked point requiring an upper cap. All fillers lie below ordinary strip tops; its ordinary copy remains top and its target parent cap is absent too.

Root parent. The ordinary left foot's effective image is R_λ . There is no anchor-column point before the next real root cap. For a non-highest cut this follows from the definition of R_λ ; for the highest cut use the argument about the actual root cap V^* in the proof of Proposition 3.6, not the artificial cut height σ . The upper tail is retained exactly by grafting. The target parent cap is therefore the corresponding real root cap in the anchor column, and Lemma 6.1 gives its exact word. Absence of a cap is preserved as well.

Good parent. The left foot and parent cap are unchanged, and every address in their profile lies below c .

Induct on target columns from left to right. Ordinary dimensions are unchanged, and these three cases identify parent-cap words exactly. Normalization, padding and trailing-coordinate replacement commute with strictly increasing address relabeling. Every ordinary non-bottom copy consequently satisfies

$$I(\text{copy}(e)) = \tau(I(e)). \quad (6.1)$$

Each filler shares the left-foot column of its strip top and lies below that ordinary top. Reconstruction Theorem 4.8 makes the target canonical, so Theorem 5.5 makes the filler word strictly smaller. Every source node has an ordinary copy. Thus fillers neither introduce new parents nor increase a parent's greatest word. The complete highest-profile column is exactly the address-relocated source table.

7. Indexed fundamental-sequence commutation

Let $G = \Phi(s) = (C_0, \dots, C_j)$, c the controlling parent, $w = j - c$, and D the lowered column of (5.2). The repaired source keeps $C_0, \dots, C_{j-1}$ and replaces its last column by D .

By Section 6, the long expansion's compact table is the retained prefix followed by

$$D, \phi_1(C_{c+1}), \dots, \phi_1(C_{j-1}), \phi_1(D), \dots, \phi_n(C_{c+1}), \dots, \phi_n(C_{j-1}), \phi_n(D).$$

For $n = 0$ only D is appended. Default FS removes the final $\phi_n(D)$, leaving

$$C_0, \dots, C_{j-1}, \prod_{b=0}^{n-1} (\phi_b(D), \phi_{b+1}(C_{c+1}), \dots, \phi_{b+1}(C_{j-1})). \quad (7.1)$$

The product denotes concatenation, and ϕ moves parents and every word address. This is precisely `cwy_direct.mjs`. Zero, successor deletion and $n = 0$ agree by definition. This proves the first identity in (0.1).

The long convention keeps the final $\phi_n(D)$, so it commutes as well. The first positive $\ln z - 1$ term is the retained prefix followed by D , with later indexing determined solely by whether $w = 1$. Width and prefixes are preserved; hence that convention also commutes termwise.

8. Order, injectivity and standard-domain surjectivity

8.1 Distinct numeric strings have distinct tables

Fix an original left prefix u and compare final values $a > 1$ and $a - 1$. Lemma 2.3 of [W] preserves the nodes below the old top and only adds grafted rows with parent below the controlling p . The old top word strictly exceeds every retained lower word. In the decreasing-parent list, rows with parent above p stay fixed; at p the word decreases or the row disappears; new smaller-parent rows cannot reverse that first difference. Therefore

$$\Phi(u, a - 1)_{\text{last}} < \Phi(u, a)_{\text{last}}.$$

Finite induction on final values, and left-locality of mountain construction, give for any raw source strings

$$s <_{\text{lex}} t \iff \Phi(s) <_{\text{collex}} \Phi(t). \quad (8.1)$$

This proves injectivity and both directions of order preservation. Whole-column prefixes are preserved exactly. On the image, an inverse may enumerate positive integers column by column until each target column matches. Injectivity gives uniqueness and membership in the image gives termination. This is computable invertibility, not an efficiency claim; the expander does not use such enumeration.

8.2 The image is the whole corresponding standard system

Let Std_{wY} be the finite closure of seeds $(1, m)$ under chosen expansion operations and prefixes. Define Std_{CWY2} similarly from the corresponding E_m .

Seeds correspond, Section 7 gives operation commutation, and prefixes commute. For the seed assertion itself, in $(1, m)$ every left foot of column 1 belongs to column 0. Repeated subtraction of value 1 creates dimensions $0, 1, \dots, m - 2$, ending with word $(0, 1, \dots, 1)$ containing $m - 2$ copies of 1. When $m = 1$ the second column is empty, giving E_1 .

Induction on finite generation in each direction now yields

$$\Phi(\text{Std}_{wY}) \subseteq \text{Std}_{CWY2}, \quad \text{Std}_{CWY2} \subseteq \Phi(\text{Std}_{wY}).$$

Together with (8.1), this is an onto order isomorphism. Corresponding terms have the same ordinal position, not merely a lower bound. Extend $\Phi(\text{Limit}_{wY}) = \text{Limit}_{CWY2}$; their n th terms both correspond to E_{n+1} , so the result extends to the external tops.

Zero corresponds to the empty string. A last column is empty exactly when its source last value is 1, so successor detection and predecessor deletion correspond as well.

Scope restriction: none of this says that every free list accepted by the NER structural parser comes from wY . Structural validation is not canonical-image or standard-domain membership. Nor is the lexicographic order on all raw positive-integer strings claimed to be well-founded.

9. Well-ordering and axiom strength

The equivalence argument uses finite mountains, address words, column induction and finite-generation induction. It does not first assume global wY well-ordering and adds no large-cardinal principle. Accepting the main wY well-ordering result of [W], the standard-domain theorem transfers through the isomorphism in its $KP_\omega +$ there exists an uncountable ordinal framework.

No exact weakest arithmetic theory for this equivalence is identified. There is no Lean formalization here. A paper derivation and kernel verification remain distinct deliverables.

10. Historical diagnostics, not proof premises

The original `test_equivalence_lemmas.mjs` tested the intermediate objects, not just final output agreement. Its fixed examples and random raw inputs covered 356 strings, with no reference-engine resource skips:

- 90,658 origin queries matched actual right-foot profiles.
- 40,833 slice-lifting checks covered equal height, dimension, lifted word and parent domination.
- 86,832 positive-dimensional intermediate-level checks included 973 changed-parent cases.
- 90,658 original-top deletions and 90,658 full-parent-merge versus actual-tail comparisons agreed.

That historical run took about 2 seconds and 216 MiB RSS. It had a 15-second deadline, a 512 MiB memory limit and a 384 MiB Node heap. It used no enormous last-value countdown or resident background process. These diagnostics can detect transcription and boundary mistakes; Sections 2-8, not the counts, are the mathematical argument. The private research harness is not a dependency of this public proof.

11. Correspondence with the code

Function or step	Paper justification
<code>belowOrigin</code>	Intermediate-level bound in Section 3; group scan and zero-dimensional inheritance in Section 4
<code>removeOriginalTop</code>	Equation (4.2) and Section 5.1
<code>minusColumn</code>	Section 5.2 and equation (5.2)
<code>fs</code> blocks and address shifts	Exact copies in Section 6; Section 7
<code>compare</code>	Section 8.1
seed and NER Limit	Section 8.2
<code>FS_alter</code> , <code>FS_short</code>	End of Section 7

12. Repository edition and sources

This is the 2026-09-17 repository edition of the existing 2026-09-16 argument, not an additional Lean proof. Section 10 describes its original historical diagnostics, not a claim that this packaging run repeated them. Current release checks are in `VALIDATION`.

The rules and readable core are included. Historical “equivalence pending” comments in the frozen code are superseded by this paper’s status, not by a machine certificate.

[W]: *Well-foundedness of the Weak-Magma omega-Y System in KP with an Uncountable Ordinal*, supplied privately as `omega-Y-Weak-Magma-KP.pdf`. The dependency list in Section 0 states which results are used. The original PDF is not redistributed and its complete well-ordering argument was not newly independently audited in this packaging task. Source and licensing boundaries are in `SOURCES`.